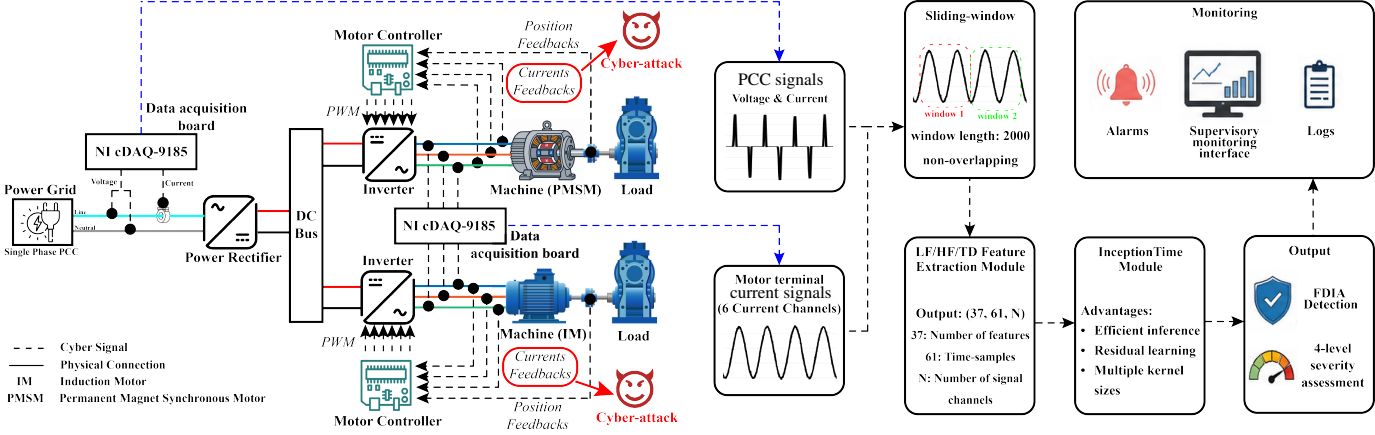

Motor Controller
Position Feedbacks
Currents Feedbacks
Cyber-attack
PWM
Data acquisition board
NI cDAQ-9185
Power Grid
Voltage
Current
Line
Neutral
Single Phase PCC
Power Rectifier
DC Bus
Inverter
Machine (PMSM)
Load
NI cDAQ-9185
Data acquisition board
Inverter
Machine (IM)
Load
PWM
Currents Feedbacks
Position Feedbacks
Cyber-attack
Motor Controller
Cyber Signal
Physical Connection
IM Induction Motor
PMSM Permanent Magnet Synchronous Motor
PCC signals
Voltage & Current
Motor terminal current signals
(6 Current Channels)
Sliding-window
window 1
window 2
window length: 2000
non-overlapping
LF/HF/TD Feature Extraction Module
Output: (37, 61, N)
37: Number of features
61: Time-samples
N: Number of signal channels
InceptionTime Module
Advantages:
• Efficient inference
• Residual learning
• Multiple kernel sizes
Monitoring
Alarms
Supervisory monitoring interface
Logs
Output
FDIA Detection
4-level severity assessment